00 / 12D — MASTER ARCHITECTURE INDEX & MULTI-WAVE AI PATCH ORCHESTRATION PROTOCOL

Complete 69-Document Knowledge-Base Index, Cross-Module Coupling Guardrails & Conflict-Free Execution Blueprint for Autonomous **GPT** & **Gemini** Developers (`JianghuSect 2 JS2-v2.6.0` $\to$ `v2.7.0+`)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Knowledge Base Folder**: `คลังข้อมูลjs2` (`69` comprehensive engineering audit specifications)
- **Primary AI Consumers**: **GPT** & **Gemini** Autonomous Development Agents
- **Entry-Point Rule**: **READ THIS DOCUMENT FIRST BEFORE PLANNING OR EDITING ANY FILE IN `GAME_SOURCE/`.**

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1. The 7 Non-Negotiable Constitutional Laws of `JianghuSect 2`

Every patch generated by **GPT** or **Gemini** MUST uphold all 7 laws without exception:

1. **Byte-Identical `human_core.js` Lock**: **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`, SHA-256 `a0d393284fb4b136848e5b80e12f3edd2c4f9437c419c793c2a7e75d881c0542`). Fix core trait/aging/stat issues strictly via `data/data_core.js` and external wrapper hooks on `g.HC` (`e.g., HC.addTrait, HC.traitMods, HC.ageFactor` in `js/cultivation.js`).
2. **100% Emergent NPC Autonomy (`Zero Mandatory Player Micromanagement`)**: Every system in the world (postures, meridians, tribulations, charters, 6-officer councils, peak policies, tribunials, sieges, bounties, espionage, marriages, births) MUST happen organically via NPC personality, epistemic knowledge, and utility AI. **The player is an Observer + Sandbox God-Editor (`UI.rEdit` stat/state tweaks & item spawning only)—never an RTS commander.**
3. **Strict Closed-Loop Resource & Spirit-Stone Conservation (`Debit === Credit`)**: Outside explicit Player Sandbox Spawns (`tracked in Periodic Digest Bucket`

#4), 0 Spirit Stones, pills, or materials may be conjured from thin air or deleted into black holes (route 100% of stone movements through ECO.transferStones).

4. **Exact 40-<script> Tag Architecture & Offline Single-HTML Parity**: Do not add or reorder <script> tags in index.html or test/load.js; always rebuild ITEM_ART_CATALOG (if modified) via node tools/build_item_art.js and bundle via node tools/build_single.js.
5. **Read-Only Zero-RNG / Zero-Mutation UI & Canvas Rendering**: ui.js view builders, render.js draw methods, and AI can()/score() evaluators must NEVER advance W.rng or mutate serialized world state.
6. **Performance Budget**: Maintain <= 3.5 ms/tick at 80 living NPCs (currently ~0.7–1.8 ms/tick).
7. **Zero Audio Scope**: Do **not** spend tokens or bundle size expanding audio/sound (SFX is intentionally out-of-scope).

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2. Complete 69-Document Directory (คลังข้อมูลjs2)

Tier 0 — Initial Macro Overview (1 Doc)

- JS2_v2.6.0_Logic_and_Contextual_Audit_for_AI: Executive 23-bug cross-system overview.

Category 01 — All 43 Cultivation Paths & 286 Subpaths (8 Docs: 01A–01H)

- **01A**: Lifecycle, Mortal Initiation, SECT.PATH_LIBRARY Mute-Grandmaster Paradox, Path Selection (×62 MAX_CAP bug), Subpath Aliases & Retraining Lifespan Trap.
- **01B**: 4/5/8/9-Stage Math Curve (GOD 38.5× XP shortcut, effectiveTechStage Stage-1 bypass), 80 Unused Peak Stations, Dormant Era-5 Postures & PATH_DESCRIPTORS.
- **01C**: 5 Founding Pillars (XIAN, BODY, GOD, DEMON, HEART_SOUL — 65 subpaths, copy-pasted realmBuffs & metaphorical skillGate bugs).
- **01D**: 7 Specialist Paths (MECH_PUPPET 0 || 100 & 'OVERDRIVE' bug, KARMA_SAMSARA 'A' cache collision, BEAST_SOVEREIGN, SAGE_SUPPORT, CRAFT_CREATION, ARRAY_TALISMAN, STRATEGY_GOVERNANCE — 67 subpaths).
- **01E**: 8 Cosmic/Dharma Grand Dao Paths (CELESTIAL_WEATHER, ASTRO_DIVINATION, VOID_SPACE, CHRONO_EPOCH, GRAVITY_DAO, PRIMORDIAL_CHAOS, BUDDHA_DHARMA, DRAGON_VEIN_GEOMANCY — 48 subpaths + stats.js:503 ternary bug).
- **01F**: 8 Scholarly/Worldly/Esoteric Paths (MERCANTILE_DAO, INK_LITERATI, LORE_ARCHIVIST, WANDERING_DAO, SONIC_HARMONY, VENOM_GU, SOUL_PUPPETRY self-link bug, PRIMORDIAL_SHIFTER — 46 subpaths).

- **01G**: 8 Martial/Folk Grand Dao Paths (SHADOW_SAMURAI 0-Qi exploits, NINPO_SHADOW, NINPO_ELEMENTAL, ARENA_GLORY, FORTUNE_GAMBLER, TIDE_VOYAGER, SPIRIT_HUSBANDRY, ANCESTRAL_RITUAL — 32 subpaths).
- **01H**: 7 Higher-Realm/Occult/Abyssal Paths (DIVINE_SUMMONER/POSSESSION/CHANNELER **deity.atkMult - 1 = -76% P0 bug**, OCCULT_MAGE missing 'ironSkin', VOID_HERETIC_PACT, DEVOURING_FLESH eating rice/stones, NETHER_GRAVE — 28 subpaths).

Category 02 — NPC Brain, Epistemic Knowledge & Autonomous Decision-Making (3 Docs: 02A–02C)

- **02A**: 4-Tier Epistemic Knowledge Matrix (Common Sense vs Hub-Synced vs Domain Literacy vs Private Info), .slice(0,24) library blindness, telepathic wallet/fog scans & social knowledge sharing.
- **02B**: Orphaned Systems (Pagoda locked at floor 9/43, 4th Sanctum skipped, Micro-Locations/Apex Trials 0% AI), _cogModCtx 59-min raid blindness, pickBasic RNG impurity & unread W.sect.petitions.
- **02C**: Executive Autonomy (Sect Master, 6-Officer Council, Outposts), Combat Archetype tactics typo (60% monks/tanks classified as 'CUNNING_ASSASSIN'), & in {} dead ternaries in expedition.js / sect.js.

Category 03 — Martial Arts & All 7 Technique Categories (7 Docs: 03A–03G)

- **03A**: Combat Moves (54 mis-tagged ART190_* dead techniques, _skListCache mastery blindness, chooseSkill 0.62 vs 1.15 threshold starving sword skills, 0 mastery from practice/spar, new weapon 0.65× mismatches & TECH_GCD).
- **03B**: Composable Engine, 34 Elemental Fusion Reactions (**'DARK_YIN' vs 'YIN_SHADOW' killing all 6 Yin fusions**), reflectOf bypasses & Duet skills.
- **03C**: Mantras (122+ sutras, EVERGREEN +63% AoE/DoT heal, MANTRA cross-map DoT reflect, Grand Formation helping traitors in grudge/leadership, & bestMantra instant old-age death trap).
- **03D**: Qinggong, Sword/Cloud Flight, Aerial MAP.blocked stranding, MOVEMENT_STEALTH_RECON 0% recon effect, & useSkill early return breaking Archer retreat.
- **03E**: Defensive Kungfu (64 arts 0% autonomous, DoT feeding free Taiji Qi, 7-layer cheat-death stacking), 3-slot Passive vs Profession conflict, & 250/286 Innate arts excluded from Constellations.
- **03F**: Support, Healing (pre-damage heal order bug, missing maxHealR on lowestAlly), Tribulation Bodyguarding (護法), Distress Flares, & 64 PROFESSION_LIFESTYLE manuals with 0% crafting/farming hooks.
- **03G**: Forbidden Arts (0 real lifespan deduction on *"sacrifice 50 years"* arts, STRICT_ORTHODOX_BAN unenforced, self-bleed/nova firing in friendly tourney/spar).

Category 04 — All 16 Weapon Families & Holistic Weapon Ecosystem (5 Docs: 04A–04E)

- **04A**: SWORD (Heavy Sword typed Swift, Yitian guardUp miss), BLADE (Xiaoli Flying Dagger broken on 0-Qi Samurai & forced melee), SPEAR, STAFF (Dog-Beating Staff copy-pasted onto 3 Legendary Armors).
- **04B**: FIST ("กรงเล็บ" name collision & -16% base stat deficit), CLAW, BOW (Condor Bow clue missing skill: 'BOW'), NEEDLE (GOLDEN_NEEDLE_SAINT disliking NEEDLE!), WHIP.
- **04C**: FOCUS ("พัดหยก" collision), ZITHER, BRUSH (Chunqiu Brush applying physical armorbreak instead of mdefbreak), ROSARY (KARMA_SAMSARA missing ROSARY), & weasel-hair/python-tendon polluting Mines/Herb Gardens.
- **04D**: FAN (Chinese char in nameTh), ASTROLABE, ABACUS — **all 3 Legendary statuses (tempoBreak, gravityCrush, contractDefault) have 0% runtime stat/combat effect**!
- **04E**: Holistic Weapon Ecosystem — 0-ms merchant buy-then-dump deletion, EQ-02 rabbit-farming T7 evolution exploit, 20 VC vs 31 REL_V230_* split, SOUL_FORGE autonomy, & socketElementalRunes burning MANUAL_PATH_*!

Category 05 — Non-Weapon Equipment, Procedural Engine & Master Synthesis (7 Docs: 05A–05G)

- **05A**: Body Armor (HEAVY_ARMOR/LIGHT_ARMOR/HEADGEAR tag mismatch, 3 Legendary Armors copy-paste bug, permanent adaptiveState on T1 cloth, applyArmorArtisan wiped by runes).
- **05B**: Jewelry & Wearables (4 Legendary Accessories bugs, PENDANT/ORNAMENT/SHOES tag exclusions, flat 1.0 AI suitability, dead EFX_SUPPLY_EFFICIENCY & EFX_QI_RESERVOIR).
- **05C**: 24 Wearable Talismans (def.charges vs def.maxCharges typo, stock vs gear trap mismatch, immortal paper talismans), 12 Formation Cores, & 0%-effect Crafting/Herb Tools (herbYieldBonus unread).
- **05D**: 62 Resonance Sets (wrong-element match bypass, missing Legendary wildcard), one-way E03 Paired element check, 3★ Relic -81.8 ATK loss on Save/Load (_vcNorm), & dormant E04 Curses.
- **05E**: Non-Weapon Overview (ACCESSORY 9-category slot bottleneck, **it.combatMult & it.defMult 100% unread in stats.js**, T1-cloth halting high-tier forge demand).
- **05F**: Procedural Engine (18/23 templates missing 5 recipes, 475 orphaned expansion recipes, MAT_V230_* unread materialMods, matched[0] set-hijacking, & harsh it.dur / 100 penalizing freshly repaired 40/40 gear by -60%).
- **05G**: Master 898-Item Synthesis (mintGear 4-layer wrapper onion bug, donate stealing 'gift-worthy' heirlooms, dur=0 shop scrap, & exiled NPC locking unique Legendary Relics).

Category 06 — Pills, Alchemy, Multi-Phase Elixirs & Toxicity (6 Docs: 06A–06F)

- **06A**: Basic/Recovery Pills (`qiRestore` `100×` unit mismatch, forced welfare/hourly pills shattering ascetic vows, `'P:HEAL'` ghost key in `sect.js`).
- **06B**: Growth/Longevity Pills (`20+` `EFX_*` dropped by `pillInfo`, `_dantianBonusPct` unread & unsaved, split lifespan fields, `"starter": true` on undecoded/heritage T7 formulas).
- **06C**: Breakthrough Pills (`31/43` paths missing `P:BT_*` & `BT.TREASURES`, personal bag bypassing `btTaken` resistance & wrong-path toxicity, permanent `_btBonusPct` leak).
- **06D**: Combat/Tactical/Forbidden Pills (`professionMeal` buff blocking `100%` of `ST.preFight` combat pills, free sect stock pills burned in friendly spars, `0%` AI use of `grantBuff`/`restoreBeastFatigue`, `ITEM190_PILL_012` missing `4`-year lifespan cost).
- **06E**: 48 Multi-Phase & Elemental Fusion Pills (**`pendingPillPhases` vs `multiPhasePills` typo killing `100%` of NPC Phase-2 effects**, double Phase-1 payout, swapped `ST.addStatus` duration arg, `0`-min T7 brew from T1 grass, & burning T7 herbs first).
- **06F**: Holistic Pill Overview (Unified `consumePillCore`, `applyQualityPill` ignoring `q` on `80%` of effects & toxicity, `'P:' + rid` vs `outItemId` stock split, & blind `EXT.pillNeed`).

Category 07 — Sect Governance, 25 Offices, 8 Peaks, Buildings & Politics (5 Docs: 07A–07E)

- **07A**: 25 Offices (`9` offices locked behind player button, `~680/mo` stipends vanishing into dead `officer.personalStones`, Day-0 `quartermaster` stone conjuration, `9` mismatched hall IDs).
- **07B**: 8 Sacred Peaks (**`80/144` Peak stations at `0%` usage**, split `treasuryStones` vs `vault.stones`, `0`-tick peak expeditions, Peaks stuck at `Lv.1`).
- **07C**: Buildings & Arrays (`100` housing cap vs `56` real beds, `3` virtual halls vs `4` empty `MAP.PLOTS`, `sectTotems` missing on new recruits, `5`-vs-`10` barrier element mismatch, dead guards on Leylines).
- **07D**: Politics & Law (**`TRIBUNAL` setting `detainedUntilDay` while engine checks `n.jail` — convicts walk free!**, `RECRUIT_DEFECTOR` chimeras, `INTERROGATE` T1 bandit dropping T7 manual, `'P:BASIC_HEAL_1'` outpost typo).
- **07E**: Living Sect Overview (`SIM.retire` traitors donating secret scrolls & deleting stones/pets, **`W.graveyard` string-vs-object reload crash**, `W.byId` burial desync).

Category 08 — Outside World, Geopolitics, Realms, Ecology & Clues (6 Docs: 08A–08F)

- **08A**: 19 Rival Sects (frozen at 12 legacy paths, PILLAGE/Tribute ignoring rivalSect.treasury, 0-ration sieges OUT-17, 0-tick vassal reinforce OUT-08).
- **08B**: Expeditions & v2.6.0 Chronicle (OUT-14 emergency teleport leaving ghost farmer in party, 100% instant win bounties/espionage, internal transfers polluting Digest external income, c.stage++ breaking storyThreadId).
- **08C**: Secret Realms (Pagoda floors 10–43 & 4th Sanctum 0% visited, OUT-05 Micro-Locations conjuring stones when worldPool === 0, 0-party Immortal Tomb clears).
- **08D**: Wilderness & Beasts (**CULT.menagerieTick(W) called once in SIM.create before W.sect exists and NEVER in daily/monthly loops!**, orphaned reservePets, Abyssal Dragon in "urban" terrain, beasts walking onto cliffs/water).
- **08E**: Fishing & Clues (starvation in front of cooked fish dishes, double meal deduction, dead studyMult, 6 v2.0 Legendary Weapons missing CLUE_TEMPLATES, clue fights hardcoding 'SWORD' XP, { free: true } Hermit Doctor).
- **08F**: Holistic World Overview (**Quantum Bilocation** across Outposts/Seclusion/Leylines/Exchange, visitor.stones draining W.world.resourcePool to 0 over decades, 200-year rival sect generational turnover).

Category 09 — Combat Engine & Tactical Simulation (6 Docs: 09A–09F)

- **09A**: Core Math (**dmg = C.hpMax(d) * frac making +HP inflate incoming damage 1:1**, ST.explain vs ST.get split-brain, normal hits skipping elements, dead formationTicks & allyDefAuraPct).
- **09B**: Spatial Geometry (NPC ignoring move stats & zocSlowMult, **24.0-tile telepathic ROW_BODYBLOCK**, BLACK_THREAD enemies bodyguarding each other, early return breaking Archer evade retreat, 90-tile ricochets).
- **09C**: Action Economy & Statuses (**statusTick DoTs draining poisoner's Forbidden HP & firing 25-tile Taiji/Bell laser reflects**, resolveChannelTick dropping synDmgMult * masBase, **BLOOD_HALT applying -75% healBlock to allies!**, Boss drop escrow).
- **09D**: Summons & Synergies (W.foeById summon leak, summons frozen outside raidTick, 1-HP _shadowClones tanking full AoEs, mage pets scaling off physical atk, 24 Party Synergies 0% active on-map).
- **09E**: Fight Modes & Tribulations (**CULT.TRIBULATION.tickAll in SIM.dailyUpdate running 1,440× too slow = 15 days instead of 15 minutes!**, raiders abandoning main_hall for 1 distant miner, +10 heartDemon & Taboo Nova in friendly spars, self-defense victims jailed).

- **09F**: Holistic Combat Overview (`computeFoe` `matk = 0.20*P` & `pen = 1..7` starving caster/late-game bosses, `7` uncapped reflect hooks exceeding `115%`, On-Map vs Off-Map bridge, `O(N_sect)` loops in `applyDamage`).

Category 10 — Economy, Workshops, CP & External Market (5 Docs: 10A–10E)

- **10A**: Primary Supply Chain (Xianxia `Bigu` fasting, expeditions wiping gourmet `mealSpecialties`, dual unbacked mine stone printers, `organs` omitted from `donorValue` & `A.donate`).
- **10B**: Workshops (`TALISMAN`/`ARRAY` starved in single `orders.forge` slot, raid interrupts destroying reserved Tier-5..7 inputs, `EC-10` double salvage, `ALCHEMY`-only personal craft lockout, `12–50×` craft XP deficit).
- **10C**: `CP` Monetary Engine (**`stone_grant` bypassing `c.price`, breaching `cashReserve`, breaking Digest math & siphoning to merchants**, Elder hoarding stagflation crushing Juniors `8×`, telepathic `0-CP` `guidedStudy`, midnight scroll confiscation).
- **10D**: External Market (**0-ms buy-then-dump gear deletion `-33%`**, `0%` member bargaining skill effect, `EC-06` spawned-input laundering via crafting, `EC-09` unsolicited recipe sales).
- **10E**: Holistic Closed-Loop Synthesis (`6`-wallet consolidation via atomic `ECO.transferStones`, `EC-08` Peak double-spend pill conjuration, cross-currency no-arbitrage proof).

Category 11 — Human Foundation, Psychology, Social Graph & Karma (5 Docs: 11A–11E)

- **11A**: Genome & `data_core.js` (`14`-year-old mortals rolling post-life `TR_AC_TOURNAMENT_CHAMPION` / widowed white hair at birth, `TR_BL_POISON_IMMUNE` dead stat path, blind swordsman getting flash-blinded, `NEUTRAL` constitutions missing `toughness`/`vitality`, `bioAge` senility).
- **11B**: Experience, Scars, Vows & Meridians (Dual `daoOath` vs `voluntaryVows`, **`n.meridians.autoOverdrive !== false` tautology rupturing meridians in friendly spars!**, `BODYGUARD_INTERCEPT` missing trait credit, ghost `daoScars` in `ST.rating` only).
- **11C**: Dynamic Psychology (**Inverted `+0.06 * cStage` making Stage-9 Immortals starve `66%` faster than mortals!**, `5` frozen `h.state` emotions, `4` starved `h.fulfil` values, high `heartDemon` giving `4×` peaceful enlightenment, missing `'counsel'` & 走火入魔 climax).
- **11D**: Social Graph & Lineage (**Missing consanguinity/sibling/celibacy guards in romance & marriage**, organic `spouse` disconnected from `companionId` & births, `3`-way mentorship split, `AI.kill` dropping `killer` entity).

- **11E**: Destiny & Karma (3 disconnected karma silos, idle `destiny.luck`, `RED_THREAD` FIFO eviction risk, awakening idle `human_core.js 155` stats like `smellTaste`, `thermalTolerance`, `deception` vs `readPeople`, `h.tastes`).

Category 12 — UI Agency, Canvas/VFX, Item Art, Save/Build & Master Index (5 Docs: `12A, 12B, 12B2, 12C, 12D`)

- **12A**: Player Agency Boundary (`20+` RTS command buttons moved from Observer sheets to `UI.rEdit` while preserving `UI.doAction` for tests, Edit-Hub `maxStage` clamp, `39`-path minimap colors, **exact fix for `startup_ui.js` timeout**).
- **12B**: Canvas 2D Rendering (**`31 REL_V230_*` Relics falling between `LEGENDARY_GEAR_VISUALS` and `!it.legendary` `drawT7EquipmentAura`**, `HC.bioAge` hair aging, path-authentic regalia/Qinggong visuals, Charged Skill final-tick `castFx` *(zero audio)*).
- **12B2**: Dedicated VFX & Item Art (`GOD`/`DIVINE_SUMMONER` cell collision, `LEG_V240_*` scroll icons, `"พัดหยก"`/`"กรงเล็บ"` icon-vs-`gearType` trap, Talisman jade-charm mismatch, **critical `54`-tech `[MOVE-CAT-01]` VFX co-update rule**, `2E/3E/4E` reaction scaling, `24`-slot queue priority).
- **12C**: Save/Load, Script Order & Tooling (**`cpeco.js:394` overwriting `ai.js:516` `AI.nextTechToStudy`**, **`validate_data_schema.js` skipping `data_tech.js` & `move.type === 'p'` typo**, `localStorage 5 MiB` quota safety, `build_single.js` missing `wuxia_item_atlas.png`).
- **`00 / 12D` (This Document)**: Master Index, Coupling Guardrails & 8-Wave Dual-AI Execution Protocol.

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3. Top-8 Cross-Suite Regression Traps (`READ BEFORE TOUCHING CODE!`)

1. **Trap #1 (`data_tech.js` $\leftrightarrow$ `skills.js` VFX)**: Restoring the `54` mis-tagged `ART190_*` techniques to `COMBAT_TECHNIQUE` (`03A`) **MUST** simultaneously move all `54` IDs from `PASSIVE_TO_T*_VFX` to `TECH_TO_T1..T7_VFX` (`12B2`) or `test/v260_vfx_tier_subpath_equip_suite.js` fails!
2. **Trap #2 (`render.js` Exact Key-Count Assertions)**: `test/legendary_visual_identity_suite.js:21–22` asserts `Object.keys(R.LEGENDARY_ART_VISUALS).length === 50` and `Object.keys(R.LEGENDARY_GEAR_VISUALS).length === 27`. Put `REL_V230_*` (`31` items) and `LEG_V240_*` (`2` arts) in separate auxiliary maps (`12B`)!
3. **Trap #3 (`data_tech.js` Mantra Length Assertion)**: `test/v550` asserts `JD.TECH.mantras.length === 122` (`and v260 asserts`

`MANTRA_TO_VFX_MOTIF matches`). Keep `JD.TECH.mantras.length === 122` or update `MANTRA_TO_VFX_MOTIF` + `v550` intentionally (`03C`)!

4. **Trap #4 (`ui.js` Programmatic `UI.doAction` Calls in Tests)**: Suites `v320`, `v340`, `v400`, `v530–v580` call `UI.doAction(...)` directly. Keep all `UI.doAction` handlers active and move manual override buttons into `UI.rEdit` rather than deleting handlers (`12A`)!
5. **Trap #5 (`combat.js` `ANATOMY_BREAKER` Direct Unit Test)**: `test/v340:176` calls `damageBossPart` directly without finishing a fight—only defer boss drops into `f.escrowBossDrops` when `opts && opts.fromStrike === true` (`09C`)!
6. **Trap #6 (`combat.js` `'DARK_YIN'` vs `'YIN_SHADOW'`)**: `test/v235` feeds reaction `spec.elements` directly, while real skills use `'YIN_SHADOW'`. Normalize `'DARK_YIN' -> 'YIN_SHADOW'` in both reaction definitions AND incoming aura keys so both pass (`03B`)!
7. **Trap #7 (`render.js` `ap.hasSilkRibbon === true`)**: `test/v250:166` asserts `apApex.hasSilkRibbon === true` for all 43 paths—keep the boolean `true` and vary the drawn silhouette by `sig.family` (`12B`)!
8. **Trap #8 (`40`-Script Load-Order Overwrites)**: Never redefine `AI.nextTechToStudy`, `A.breakthrough.start`, or `ECO.applyPill` in a later script without preserving/calling earlier logic (`12C`)!

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4. Conflict-Free 8-Wave Execution Plan & **GPT** + **Gemini** Division of Labor

To achieve `0` merge conflicts and `100%` green tests after every wave:

- **GPT Track (Precision Numerical Kernels, Data Schemas, Economy Ledgers & Serialization Invariants)**
- **Gemini Track (Holistic Multi-System AI Autonomy, 43-Path/Technique Semantics, Combat Tactics & Visual/Observer UX)**

Wave	Focus	GPT Assigned Scope (Disjoint Files)	Gemini Assigned Scope (Disjoint Files)	Verification Gate
Wave 1	**P0 Crashes, `-76%` Pantheon Bug, Save/Load & Validator**	`sim.js`, `tools/validate_data_schema.js`, `tools/build_`	`cultivation.js`, `data/data_core.js` (`deity.atkMu`	`core_lock`, `validate_data_schema`, `startup_ui`, `v240`

Wave	Focus	GPT Assigned Scope (Disjoint Files)	Gemini Assigned Scope (Disjoint Files)	Verification Gate
		`single.js`, `index.html` (`graveyard` crash, `W.byId`, `PERSISTED_UNDERSCORE_KEYS`, `#impFile` `startup_ui` fix, validator `data_tech` + `type==='p'` fix, `SIM.getDeploymentState` lock)	`lt -76%` P0 fix, `HC.addTrait`/`HC.ageFactor` wrappers, `TR_IN_CRIPPLED_ROOT` `MAX_CAP` fix, `TR_BL_POISON_IMMUNE`, `TR_IN_BLIND_SWORD_EAR`, `CONST_*` elements/toughness, `menagerieTick` daily hook)	
Wave 2	**Unified Stat, Equipment & Pill Kernels**	`economy.js`, `bottleneck.js` (`listedAtDay` 0-ms dump fix, `consumePillCore`, `qiRestore` scale, `multiPhasePills` queue + arg order, `applyQualityPill` full scaling, `1,370` `allRecipes`, `3` workshop queues, craft abort refund,	`stats.js`, `martial.js`, `visitors_clues.js` (Unify `ST.compute` `===` `getBaseFast`, `_hpf`, `combatMult`/`defMult`, `HEAVY_ARMOR`/`PENDANT`/`SHOES` aliases, 27 Legendary Weapon/Armor/Acc proc fixes, `tempoBreak`/`contractDefault`/`gravity_slow` runtime	`equipment_tier_art_suite`, `visitors_clues_weapons_suite`, `v350`

Wave	Focus	GPT Assigned Scope (Disjoint Files)	Gemini Assigned Scope (Disjoint Files)	Verification Gate
		`EC-10` salvage, `dur/maxDur`)	effects, 6 new Legendary `CLUE_TEMPLATES`)	
Wave 3	**Closed-Loop Economy (`ECO.transferStones`) & `CP` Engine**	`cpeco.js`, `external_market.js` (`ECO.transferStones` 6-wallet kernel, `stone_grant` `cashReserve` + `c.price` peg, `avg+median` stagflation fix, `techRetentionScore` merge in `nextTechToStudy`, `studyWaivers`, `memberTradeSkill` + `FREE_TRADE_TREATY`, `EC-06`/`EC-09`, `visitor.stones` pool recycling)	`content.js`, `fishing.js`, `wildlife.js`, `wild_resources.js`, `data/data_beasts.js` (`organs` in `donorValue`/`donate`, `A.eat` cooked-fish famine fix & single-meal deduction, `studyMult` wire, 286-subpath chest table, beast `!MAP.blocked` & aquatic terrains, `'healer'`/`'herbal'` post typo)	`cp_economy`, `external_market_expansion_suite`, `v311`
Wave 4	**Combat Engine Math, Spatial Physics, DoT & Tribulations**	`elements.js` + `combat.js` *(Math/Damage/Status half: stage-ref HP anchor `09A`, `computeFoe`*	`skills.js` + `combat.js` *(Tactical/AI/Mode half: `'YIN_SHADOW'` 6 Yin fusions, `chooseSkill`*	`combat_balance`, `v235`, `v300`, `v340`, `v530`

Wave	Focus	GPT Assigned Scope (Disjoint Files)	Gemini Assigned Scope (Disjoint Files)	Verification Gate
		caster `matk`/`pen`/`buffs`, `C.reflectOf` 7-hook consolidation, `DOT:` tag stopping self-bleed & cross-map laser reflects, `BLOOD_HALT` ally heal, `TRIBULATION` in `SIM.tick`)	*threshold/overwrite fix, `moveToward` NPC `move`+`ZoC`, `3.0`-tile `ROW_BODYBLOCK`, Archer `evade` step-back, `f.serious` guards on Meridian Overdrive/`KW_LAST_BREATH`/Taboo Nova, `raid main_hall` AI, summons in `COMBAT.tick`)*	
Wave 5	**Technique Categories & 43-Path / 286-Subpath Depth**	`data/data_tech.js`, `data/data_expansion.js`, `data/data_bt.js` (Restore `54 ART190_*` + **co-migrate `TECH_T0_T1..T7_VFX`**, fix role-inverted `progressionMapping` `realmBuffs` & metaphorical `skillGate`s, 31-path `BT` support)	`martial_paths.js`, `proficiency.js`, `retraining.js`, `knowledge.js` (`64 PROFESSION_LIFESTYLE` crafting/farming/caravan hooks, `03G` real `FORBIDDEN` lifespan cost & `STRICT_ORTHODOX_BAN` enforcement, intra-path	`v200–v230`, `v540–v580`, `v260_vfx`

Wave	Focus	GPT Assigned Scope (Disjoint Files)	Gemini Assigned Scope (Disjoint Files)	Verification Gate
			retraining stage preservation, unlocking paths via living Grandmasters)	
Wave 6	**4-Tier Epistemic NPC Brain & Full Autonomous World**	`expedition.js`, `explore.js` (Fix `in {}` dead ternaries, unlock Pagoda floors `10–43` & 4th Sanctum, autonomous Micro-Locations & Apex Trials, `OUT-05/08/12/14/16/17/18/19`, Digest `internalTransfer` reconciliation, rival sect 43-path generational scaling)	`cognition.js`, `ai.js` (4-Tier Epistemic perception replacing `.slice(0,24)` & telepathic scans, `shareKnowledgeOnInteract`, `_cogModCtx` raid invalidation, pure-RNG `pickBasic`, `Bigu` hunger fix `ai.js:972`, `5` emotions + `fulfil` loops, `'counsel'` & `走火入魔`, `canRomance` kinship/celibacy guard, autonomous Era-5 postures & vows)	`v260_living_chronicle`, `cognition` suites
Wave 7	**Living Sect Governance, 8 Peaks & Dynasty Lineage**	`sect.js` *(Fiscal/Infra/Law half: unify* `personalStones ->`	`sect.js` *(Autonomy/Peaks/Lineage half: route* `80` *Peak* `'practice'/'m`	`v310`, `v320`, `v330`, `v360`, `v400`, `v410`

Wave	Focus	GPT Assigned Scope (Disjoint Files)	Gemini Assigned Scope (Disjoint Files)	Verification Gate
		`inv.stones` & `vault.stones -> treasuryStones`, `EC-08` real stock pills, `TRIBUNAL n.jail` fix, `petitions` docket, `housingCapacity` = real beds, plot 3 virtual halls, `sectTotems` dynamic lookup)	*`editate'` stations + `peak_* Lv.2–5` upgrades, auto-unlock 9 offices by sect scale, autonomous Charter/Council/Outposts/Bounties/Espionage, unified `MENTORSHIP.formBond`, `AI.kill { killer }` grudge/threads, monthly `spawnCompanionOffspring`)*	
Wave 8	**Observer UI Purity, `REL_V230_*` Visuals & Full CI Sweep**	`docs/ITEM_ART_CATALOG.json`, `ui.js` *(Item Art & Edit Hub: fix collisions, `"พัดหยก"/"กรงเล็บ"` baseForms, Talisman/Specialty-Pill routing, `Edit Hub maxStage` clamp, move RTS buttons to `UI.rEdit` keeping*	`render.js`, `main.js` *(`R.GRAND_DAO_RELIC_VISUALS` for `31 REL_V230_*`, `LEG_V240_*` Apex scenes, `2E/3E/4E` reaction & `43`-path finisher VFX, Domain rings, `24`-slot queue priority, `HC.bioAge` hair aging, `43`-path*	**100% Full Suite (`162/162` PASS, `0` timeouts, `<= 3.5 ms/tick`)**

Wave	Focus	GPT Assigned Scope (Disjoint Files)	Gemini Assigned Scope (Disjoint Files)	Verification Gate
		`UI.doAction` intact)	*minimap colors, `0×` pause idle render skip)*	